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Where the Time Goes

Analysis of a Public LLM Serving System

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MOTIVATION

Why characterize a real LLM workload?

LLM Serving is critical day-to-day infrastructure
We need better *scheduling, autoscaling, resource allocations*

The disconnect

Current studies mostly rely on **synthetic benchmarks**

Production workloads are ***unknown***

Result: *misallocated optimization effort.*

Public production traces lack:

- phase-level timing breakdown
- per-model distributions
- cache-hit signals
- sufficient time coverage

Our contribution

1. **Five-month production trace**
   from a public, open-source serving system
2. **Workload characterization**
   identifying where latency originates in the pipeline
3. **Concrete recommendations**
   grounded in observed bottlenecks, not assumptions

THE DATASET

Overview

Pipeline from Public AI: A nonprofit, open-source service to make public and sovereign AI models more accessible

Users → **Open WebUI / API Calls** (chat / sessions) → **LiteLLM** (gateway · trace logged here) → **vLLM backends** (CSCS and other providers)

Apertus: a fully open LLM, developed by EPFL, ETH Zurich and CSCS as part of the Swiss AI Initiative

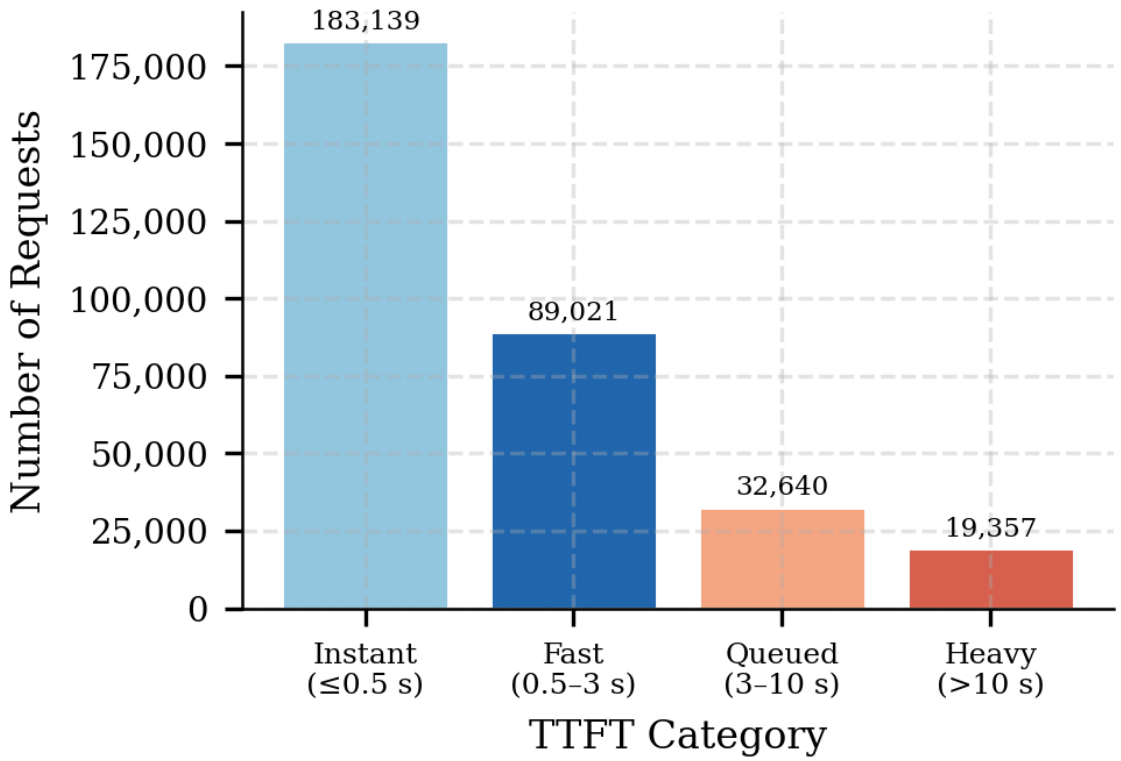
337K	5	Apertus	78 / 22
successful requests	months (Sep '25 – Jan '26)	2 variants (8B and 70B)	% traffic split (8B / 70B)

Per request: *timestamps (start/completion start/end) · tokens (prompt/completion/total) · cache-hit · turns*

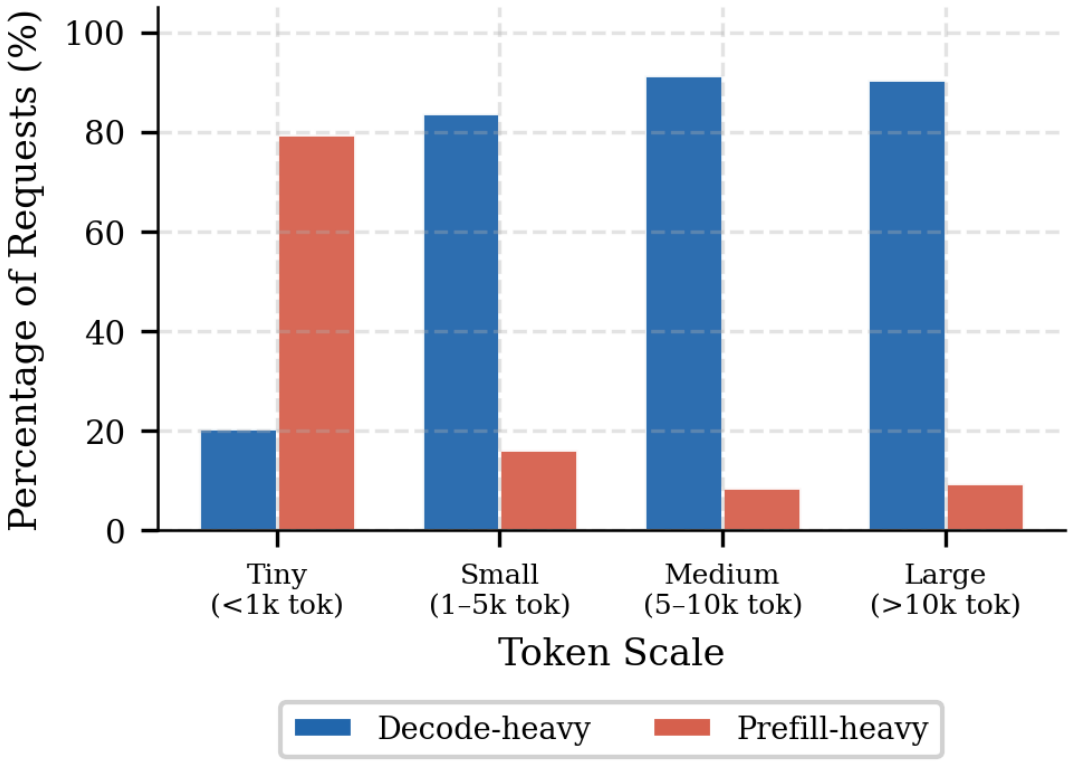
Trace release: anonymized trace + open Apertus weights + open-source framework constitutes an end-to-end pipeline.

FINDING 1

TTFT is dominated by queueing, not prefill compute



(a) Requests by TTFT category, 1 in 6 substantially queued

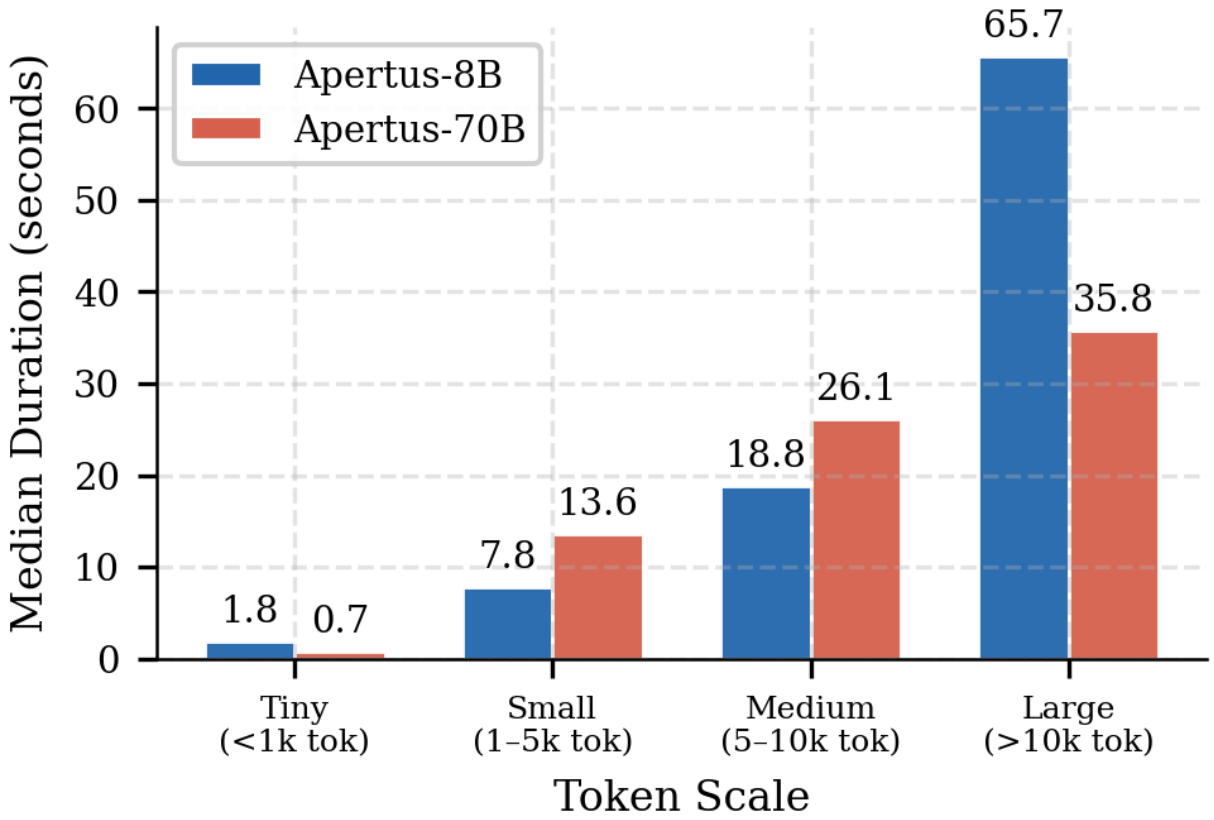


(b) Phase split by token scale, 80% of tiny requests are prefill-heavy

Tiny requests should compute in milliseconds, yet many start in seconds. **TTFT ≠ prefill.**

FINDING 2

Model size does not translate into user-perceived latency



Median end-to-end duration by model and token scale

What the user sees

For tiny and large requests, the 70B is actually **faster** than the 8B.

Compute is not irrelevant but dominated by system level factors;

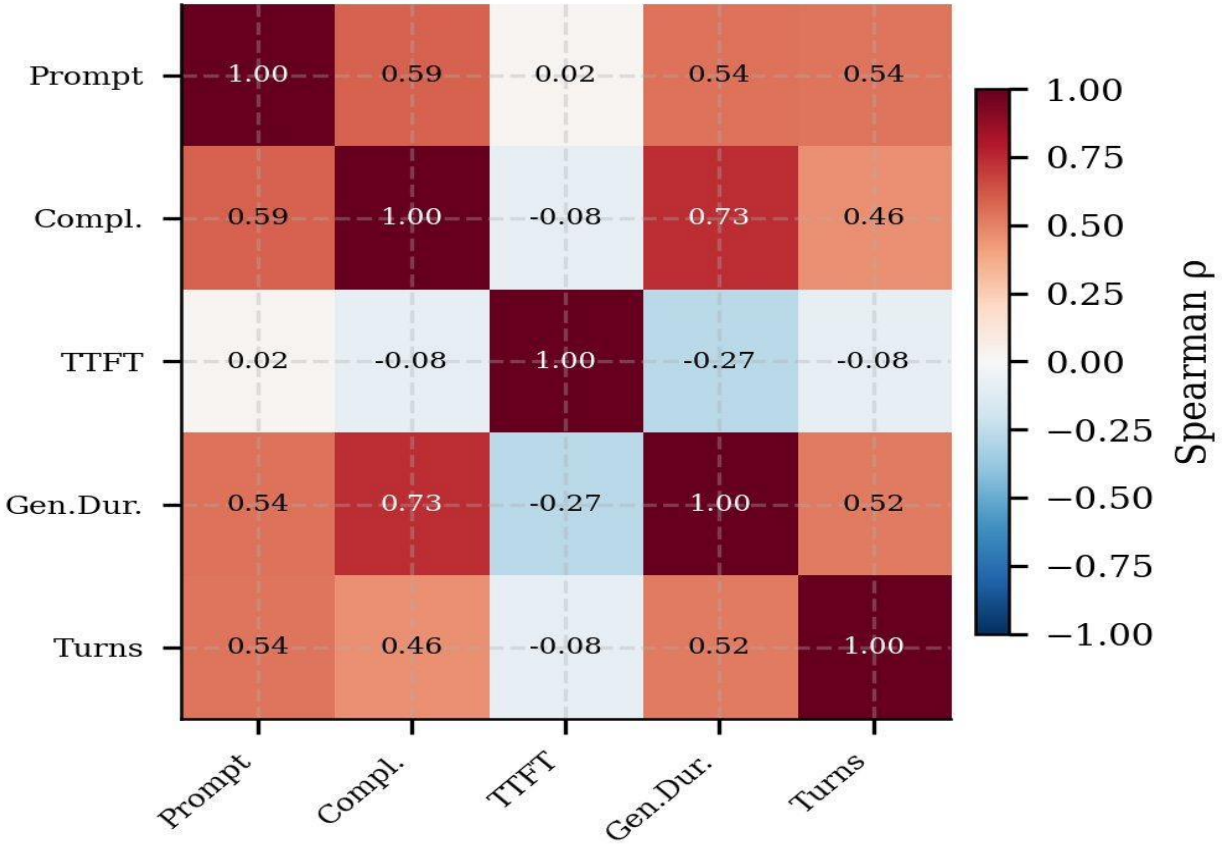
- Queuing
- Resource allocation
- Traffic

The compute advantage is consumed by queueing. (%78 to 8B, %22 to 70B)

In this stack, system-level effects mediate model size, routing on size as a latency proxy is unreliable.

FINDING 3

Arrival time features are weak predictors



The heatmap shows the correlation between the features

- *Each cell shows the correlation between two features.*
- *Darker red / blue = stronger positive/negative relationship.*
- *Near-white ≈ no relationship.*

The signals a scheduler can actually use

0.02	**Prompt Tokens ↔ TTFT** *essentially zero*
0.54	**Prompt Tokens ↔ Generation Duration** *moderate, noisy*
0.52	**Conversation Turns ↔ Generation Duration** *moderate, noisy*
0.73	**Completion Tokens ↔ Generation Duration** *but unobservable at arrival*

Point-estimate schedulers operate on weak signal, predictive policies need misprediction-aware fallbacks.

IMPLICATIONS

Treat queueing as a first-class bottleneck

1

Size-aware priority

Prioritize tiny / small requests, most of their latency is removable queueing.

2

Model-agnostic decisions

Route on observed bottleneck regime, not on parameter count as a latency proxy.

3

Dynamic capacity

Allocate along the daily traffic profile to flatten queue formation at peak hours.

4

Adaptive batching

React to live queue depth and request mix, avoid throughput-only policies that punish small requests.

Common theme: *for a large fraction of requests, time waiting > time computing.*

THANK YOU · TAKEAWAYS

Thank you for listening!

01

TTFT is queueing.

Not prefill computing. Tiny requests wait seconds for milliseconds of work.

02

Model size ≠ user latency.

Smaller, simpler model is slower in this setup; system-level effects dominate the parameter gap.

03

Predict with caution.

Arrival features are weakly or moderately correlated with duration. Schedulers need misprediction-aware fallbacks.

Trace: https://github.com/heigvd-software-engineering/apertus-serving-analysis

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